



# PPA Speech Therapy Suite

## Therapist & Caregiver User Guide

Designed for people with Primary Progressive Aphasia (PPA)  
and their speech-language pathologists and caregivers.

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## Introduction

The **PPA Speech Therapy Suite** is a browser-based tool designed to help people with **Primary Progressive Aphasia (PPA)** — a neurological condition that gradually affects the ability to speak, read, and understand language — maintain and practise communication skills.

The suite provides nine structured practice modules, plus an AI therapist assistant (Dr. Aria, powered by Claude), progress tracking, and export/import tools for therapist-customised content.

### Nine Modules at a Glance

| Module                  | Purpose                                                   |
|-------------------------|-----------------------------------------------------------|
| AI Therapist (Dr. Aria) | Free conversation and encouragement with Claude AI        |
| Assessment              | Structured language evaluation tasks                      |
| Naming Practice         | Picture/word naming with spaced repetition and cue ladder |
| Repetition              | Repeat words and phrases at increasing difficulty         |
| Script Training         | Practise functional scripts for daily life                |
| Sentence Builder        | Drag-and-drop sentence construction                       |
| Sentence Work           | Sentence completion and construction prompts              |
| Video Questions         | Comprehension questions on short video clips              |
| Progress                | Session history, accuracy trends, and SR statistics       |

### Who is this guide for?

This guide is written for **speech-language pathologists (SLPs)** and **caregivers** who set up and supervise sessions. The interface is also designed to be used directly by the person with PPA, with or without a helper present.

Sessions can be as short as 10–15 minutes. The app saves progress automatically in the browser — no account or internet connection is required for most features (an internet connection is only needed for the AI Therapist and for AI feedback within modules).

## Installation

The app is distributed as a zip archive containing an installer script, the application bundle, and this documentation. Unzip the archive, then follow the steps for your operating system below. All files must remain in the same folder.

### macOS

**Requirements:** macOS 12 Monterey or later · Node.js 18+ (the installer will guide you if missing) · Internet connection (for npm packages during install).

1. Double-click **Install PPA Therapy.command**.

If macOS shows a security warning: right-click the file → **Open** → **Open** to approve it once.

2. Follow the on-screen prompts to choose an install location and optionally enter your Anthropic API key. You can skip the key — the app will prompt for it on first launch.

3. When finished, double-click **Launch PPA Therapy** on your Desktop. The app opens automatically in your browser.

### Windows

**Requirements:** Windows 10 or 11 · Node.js 18+ (the installer will guide you if missing) · Internet connection (for npm packages during install).

1. Double-click **Install PPA Therapy.bat**.

If Windows SmartScreen shows a warning: click **More info** → **Run anyway**. The script only creates a local project — it does not modify system settings.

2. Follow the on-screen prompts to choose an install location and optionally enter your Anthropic API key. You can skip the key — the app will prompt for it on first launch.

3. When finished, double-click **Launch PPA Therapy** on your Desktop. A console window opens briefly and your browser loads the app automatically.

### Updating

To install a newer version, run the installer again with the updated bundle and choose the same install folder. Your session data and custom content are stored in the browser and are not affected. Download a backup from **Progress** → **Settings** before updating as a precaution.

### Uninstalling

**macOS:** Delete the install folder (default: `~/PPA Therapy`) and the Desktop launcher.

**Windows:** Use *Start* → *PPA Therapy* → *Uninstall PPA Therapy*, or delete `%LOCALAPPDATA%\PPA Therapy` and the Desktop shortcut manually.

## Getting Started

### Starting the App

Double-click **Launch PPA Therapy** on your Desktop. A small console window opens (this is normal — it runs the local server), and the app loads automatically in your browser within a few seconds. The recommended screen size is a tablet or laptop (10" or larger).

To stop the app, close the console window. The browser tab can be closed at any time — your progress is saved automatically.

**iPad / direct browser use:** The app can also be opened directly in Safari or Chrome without the installer — contact your clinician for the URL. On iPad, tap the Share button in Safari then **Add to Home Screen** to install it as a PWA and open it full-screen without the browser toolbar.

### First-Time Setup (SLP / Admin)

The first time you open the app, everything is ready to use with default content. To customise the word lists and materials:

- Tap **Admin** in the sidebar (PIN: 1234 — change this before clinical use).
- In the Naming module admin panel, add, edit, or remove picture cards.
- In other modules, tap the admin toolbar to edit word banks and prompts.
- Use **Export / Import** to save customised content as .ppa files that can be shared with other devices.

### Navigation

The **sidebar** on the left lists all nine modules. Tap any module name to switch. Your place within a session is preserved when you switch and return.

### API Key (AI features)

Dr. Aria and the AI feedback in each module require an **Anthropic API key**. Without a key the modules still work — they simply skip the AI feedback step and advance automatically. To add a key:

- Open Admin → Settings.
- Paste your Anthropic API key in the API Key field.
- The key is stored only in this browser — it is never sent to any server other than Anthropic.

## Module Guides

### AI Therapist (Dr. Aria)

Dr. Aria is a friendly, encouraging AI therapist powered by Claude. Use this module for open-ended conversation practice, warm-up chat, or emotional check-ins at the start of a session.

- Type a message in the box at the bottom and press **Send** (or Enter).
- Dr. Aria responds in natural, supportive language.
- There is no time limit — take as long as you need.

### Assessment

Use this module for structured language evaluation. Tasks include single-word repetition, picture description, and sentence-level prompts.

- Select a task category from the list.
- Read the prompt aloud to the patient.
- Record the response using the text box or voice notes.
- Tap **Next** to advance.
- Tap the picture to enlarge it to twice its size. Tap outside the picture or press **Escape** to close.

### Naming Practice

A picture or word card is shown. Try to name it. If the word is on the tip of your tongue, press the **Space bar** while the answer box is empty for a sound-starter hint:

- **Sound hints on (■)**: each press reveals and speaks the beginning of the word up to the next natural vowel sound, plus the following consonant — for example the first press on *glasses* shows and says **GLAS**; on *chair* it shows and says **CHAIR**. The pronunciation always matches how the word really sounds in context, not how the letters would sound in isolation.
- **Sound hints off**: each press reveals one letter group at a time with no audio.

The ■ button in the bottom-right corner of the hint area toggles sound hints. Your preference is saved automatically.

For more support, use the cue buttons:

- **Semantic Cue** — shows a written description or category clue (e.g. "a fruit").
- **Phonemic Cue** — shows the first sound of the word in square brackets (e.g. [g]).
- **Reveal** — shows the full word. Use this if no other cue has worked.

**Answer box note:** if there is already text in the box, the Space bar adds a space character rather than giving a hint — clear the box first if you want a hint.

When the patient types their answer, tap **Got it!** to record a correct response and get Dr. Aria's feedback, or use the cue buttons if more support is needed. Tap **Next word** → after Dr. Aria responds (or immediately if AI is not connected).

Tap the picture card graphic to enlarge it to twice its size — useful if the image is small or the patient needs a closer look. Tap outside the enlarged image or press **Escape** to dismiss it.

### ■ Spaced Repetition — how words are scheduled

The Naming module uses a PPA-adapted spaced repetition engine. Words you recall easily are shown less often; words that need more support are shown more often. The maximum interval is 5 days — every word is reviewed at least weekly.

#### **Result scoring:**

correct (no cues used)  $\times 1.2$  interval

space\_cued (Space-bar hint used)  $\times 0.9$  interval

semantic\_cued  $\times 0.8$  interval

phonemic\_cued  $\times 0.5$  interval

failed (full reveal) reset to 1-day interval

Words scored phonemic\_cued or failed are re-queued 4 positions ahead for a second attempt within the same session.

### ■ Admin — customising the word list

Add item: type a word; AI looks up an emoji and fills in cues.

Bulk import photos: select multiple images; AI fills empty word fields.

Generate by category: enter a category name and count; AI generates nouns.

Edit / Delete: change any field or remove cards. Export list as a .ppa file.

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## Repetition

Structured repetition drills at increasing levels of difficulty.

- Choose a difficulty level (1 = single words, up to 5 = full sentences).
  - The prompt is shown on screen.
  - The patient repeats the word or phrase aloud.
  - Tap **Correct** or **Incorrect** to record the result.
  - Tap **Next** to advance to the next item.
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## Script Training

Functional scripts are short, high-frequency phrases for daily life — ordering food, greeting someone, answering the phone. Repeated practice builds automatic retrieval.

- Select a script from the list.
  - The script is shown line by line.
  - The patient reads or repeats each line.
  - Tap **Next line** to advance.
  - Dr. Aria can provide feedback at the end of a script if AI is connected.
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## Sentence Builder

A visual drag-and-drop tool for constructing sentences. Word tiles are arranged into categories (nouns, verbs, adjectives, etc.) and dragged into a sentence tray.

- Tap or drag a word tile from the word bank into the sentence tray.
- Rearrange tiles by dragging them within the tray.
- Tap **Check sentence** to get AI feedback.
- Tap **Clear** to start again.
- Filter the word bank by category using the tabs at the top.

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## Sentence Work

Two sub-modes for sentence-level language practice:

**Sentence Completion** — a sentence with a missing word is shown. The patient types or says the missing word.

**Sentence Construction** — a topic or picture prompt is shown. The patient constructs a complete sentence.

After each response, Dr. Aria gives spoken and written feedback if AI is connected. Tap **Next** to advance.

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## Video Questions

Short video clips (from YouTube) are played, followed by comprehension questions. This module supports auditory and visual processing alongside language.

- Select a video from the list.
- Watch the clip (plays embedded in the app).
- Answer comprehension questions — multiple choice or open-ended.
- Tap **Submit** to record your answer.
- Tap **Next question** to continue.



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## Progress

The Progress module shows a summary of all recorded activity.

**Session log** — a chronological list of activities from the current session.

**Accuracy trends** — correct/incorrect rates by module over time.

**Naming SR statistics** — per-word spaced repetition data: interval, streak, last result, next due date.

Progress data is stored locally in the browser. Use **Export Backup** to save a .ppabak file as an off-device copy.

## Admin Panel

The Admin Panel is protected by a 4-digit PIN (default: **1234**). Change this before clinical use by editing the source code constant `ADMIN_PIN`.

### Naming Module Admin

- **Add item** — type a word; the app will look up an emoji automatically (internet required), or you can upload an image.
- **Bulk import photos** — drag-and-drop or select multiple images at once. A grid shows each photo with fields for word, category, semantic cue, and phonemic cue. Blurring the Word field auto-fills the other three via AI. "Generate AI defaults for all" processes every row at once. Only rows with all fields filled are saved; words already in the list are skipped.
- **Generate by category** — type a category name and choose a count (default 15, max 50). The app calls the AI to generate the most common nouns in that category with emoji, semantic cue, and phonemic cue pre-filled. New items are pre-selected; duplicates are shown with their existing values and skipped automatically.
- **Edit item** — change the word, graphic, semantic cue, or phonemic cue.
- **Delete item** — removes the card from the practice list.
- **Export list** — saves the current list as a .ppa file.
- **Import list** — loads a .ppa file, merging with or replacing the current list.

### Other Module Admin

Each module with editable content shows an admin toolbar at the top of the screen when Admin mode is active. Use it to add, edit, and delete items.

#### ■■ Admin PIN security

The default PIN is 1234 and is visible in the source code. Change it before using the app in a clinical setting. The PIN only prevents accidental edits — it is not a security feature.

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## Backup & Restore

All app data is stored in the browser's **localStorage**. This means:

- Data persists between sessions on the same device and browser.
- Clearing browser data or switching browsers will lose data.
- Data does not sync across devices automatically.

### To back up:

- Open Admin → Backup & Restore.
- Tap **Export Full Backup** to download a .ppabak file.
- Store the file in a safe location (e.g. a shared folder or email it to yourself).

### To restore:

- Open Admin → Backup & Restore.
- Tap **Import Backup** and select the .ppabak file.
- The page will reload with all data restored.

■ **Tip — backup after every customisation session**

Whenever you add new picture cards, edit scripts, or adjust word banks, export a fresh .ppabak backup. This protects your work if the browser data is ever cleared.



## Tips for Effective Sessions

### Session length and pacing

Shorter, more frequent sessions are generally more effective than long infrequent ones. Aim for 15–30 minutes per day if possible. Watch for signs of fatigue — reduced engagement, increased errors, or frustration — and end the session early if needed.

### Choosing the right modules

Begin sessions with a warm-up activity (AI Therapist or a well-known Script) before moving to more demanding tasks (Naming, Sentence Work). End on a positive note with something the patient finds rewarding.

### Using cues effectively

In Naming Practice, try to allow a generous response time (10–15 seconds) before offering a cue. Use the least-support cue first — Space-bar phoneme hint, then Semantic Cue, then Phonemic Cue, then Reveal. This hierarchy is reflected in the spaced repetition scoring.

### Involving the patient in goal-setting

Where possible, involve the person with PPA in deciding which words to practise and which scripts matter most to them. Personally meaningful content produces better motivation and retention.

### Recording baseline data

Before starting intensive practice on a new word set, record a baseline probe in the Assessment module. Repeat the probe every 2–4 weeks to track maintenance and generalisation.

### Caregiver involvement

Caregivers can support practice by:

- Sitting alongside and providing gentle encouragement (not corrections).
- Reading prompts aloud if the patient finds on-screen text difficult.
- Noting which words or scripts the patient found particularly difficult or easy.
- Exporting a session log to share with the patient's SLP.

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## Troubleshooting

Below are the most common issues and how to resolve them.

| Problem                                  | Solution                                                                                                                                                        |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dr. Aria does not respond                | Check your API key in Admin → Settings. Make sure you have an internet connection. The module will still advance without a response.                            |
| Picture cards show ■ instead of an image | The emoji look-up requires an internet connection. Try again online, or upload a photo from the camera roll in Admin mode.                                      |
| Space bar does not give a hint           | Make sure the answer box is empty. If there is any text in the box the Space bar adds a space instead. Clear the box first.                                     |
| All my word lists have disappeared       | Browser localStorage may have been cleared. Restore from a .ppabak backup file in Admin → Backup & Restore.                                                     |
| The app is very slow                     | Close other browser tabs. On iPad, ensure Low Power Mode is off. The app runs best in Chrome or Safari on a modern device.                                      |
| Export / Import does not work            | File downloads may be blocked by a browser setting or pop-up blocker. Check browser permissions and try again.                                                  |
| Audio hints do not play (sound icon)     | Check that the device is not muted and the browser has permission to play audio. On iOS, the silent switch must be off. Tap the ■ button to toggle audio hints. |